CORE PURE / DIFFERENTIAL EQUATIONS

Modelling with 1st Order Differential Equations

Mark scheme 14 questions · 225 marks

Questions

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	Total	**225**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$2\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{12t-4v}{1+t}$ $\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + \frac{\ldots}{1+t}v = \ldots$	**M1**	Uses $F = ma$ with acceleration $\frac{\mathrm{d}v}{\mathrm{d}t}$ and collects the v term on the LHS. Condone a slip dividing by the mass.
	$P(t) = \frac{2}{1+t}$, $Q(t) = \frac{6t}{1+t}$	**A1**	Both correct, oe.
		[2]	
1(b)	$e^{\int \frac{2}{1+t}\,\mathrm{d}t}$	**M1**	Uses $e^{\int P\,\mathrm{d}t}$ with their $P(t)$ of the form $\frac{k}{1+t}$.
	$= e^{2\ln(1+t)} = (1+t)^2$	**A1**	
	$(1+t)^2 v = \int 6t(1+t)\,\mathrm{d}t$	**dM1**	Writes LHS as their integrating factor times v, and integrates their $Q(t)$ times it, at least one power increased by 1.
	$(1+t)^2 v = 3t^2 + 2t^3 + c$	**A1**	Constant required; oe.
	$t = 0,\ v = -5 \Rightarrow c = \ldots\ [= -5]$	**M1**	Substitutes $t = 0$, $v = -5$ into their general solution containing a constant.
	$v = \frac{2t^3+3t^2-5}{(1+t)^2}$	**A1**	oe.
	$2t^3 + 3t^2 - 5 = 0$ $\Rightarrow (t-1)(2t^2+5t+5) = 0$	**M1**	Sets their numerator equal to zero and finds a root by any means, e.g. factor theorem then factorising; a calculator root is accepted.
	$2t^2 + 5t + 5 > 0$ for $t \geq 0 \Rightarrow t = 1$	**A1**	$t = 1$ with the quadratic factor shown non-zero for $t \geq 0$ (negative discriminant or positive terms), or another argument excluding an earlier root.
		[8]	
	Total for Question 1	**10**	

Question	Scheme	Marks	Notes
2(a)	$V = 600 + 20t - 30t = 600 - 10t$	**B1**	oe, e.g. $10(60 - t)$.
	Rate in $= 20 \times 3 = 60$	**B1**	
	Rate out $= 30 \times \frac{x}{600-10t}$	**M1**	Uses outflow rate $\times$ x over their volume, where the volume depends on t. Not for $\frac{x}{600}$.
	$\frac{\mathrm{d}x}{\mathrm{d}t} = 60 - \frac{30x}{600-10t} = 60 - \frac{3x}{60-t}$	**A1 AG**	Rate in minus rate out, both correct, leading without error to the printed equation; combined simplification accepted.
		[4]	
2(b)	$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3x}{60-t} = 60$ $\Rightarrow \mu = e^{\int \frac{3}{60-t}\,\mathrm{d}t}$	**M1**	Rearranges to $\frac{\mathrm{d}x}{\mathrm{d}t} + Px = Q$ and forms $e^{\int P\,\mathrm{d}t}$ with $P = \pm\frac{3}{60-t}$. Condone a sign slip.
	$= (60 - t)^{-3}$	**A1**	oe, e.g. $\frac{1}{(60-t)^3}$.
	$\frac{x}{(60-t)^3} = \int 60(60 - t)^{-3}\,\mathrm{d}t$	**dM1**	Writes LHS as $x\times$ their integrating factor and integrates RHS, at least one power increased by 1. Condone sign and coefficient slips.
	$\frac{x}{(60-t)^3} = 30(60 - t)^{-2} + c$	**A1**	Must include the constant; oe, e.g. $x = 30(60 - t) + c(60 - t)^3$.
	$t = 0,\ x = 0 \Rightarrow c = \ldots\ [= -\frac{1}{120}]$ $t = 30 \Rightarrow x = \ldots$	**M1**	Uses $x = 0$ at $t = 0$ in their general solution with a constant to find it, then substitutes $t = 30$.
	$x = 675$	**A1**	
		[6]	
2(c)	Mixing is not instant or uniform; refine by letting the concentration vary through the tank	**E1**	States a valid limitation, such as non-instant mixing or non-constant flow rates, and a refinement addressing it. Both needed.
		[1]	
	Total for Question 2	**11**	

Question	Scheme	Marks	Notes
3(a)	$V = 600 - 3t$	**B1**	oe, eg $3(200 - t)$.
	Rate in $= 6 \times 2 = 12$	**B1**	
	Rate out $= 9 \times \dfrac{x}{600 - 3t}$	**M1**	Multiplies 9 by x divided by their volume, which must be a linear expression in t.
	$\dfrac{\mathrm{d}x}{\mathrm{d}t} = 12 - \dfrac{9x}{600 - 3t} = 12 - \dfrac{3x}{200 - t}$	**A1 AG**	Rate in minus correct rate out, leading without error to the printed equation; cancelling by 3 may be combined.
		[4]	
3(b)	$\dfrac{\mathrm{d}x}{\mathrm{d}t} + \dfrac{3}{200 - t}x = 12$ $\Rightarrow e^{\int \frac{3}{200-t}\,\mathrm{d}t} = e^{-3\ln(200-t)}$	**M1**	Attempts $e^{\int P\,\mathrm{d}t}$ with $P = \pm\frac{3}{200-t}$, integrating to the form $k\ln(200 - t)$. Condone sign slips.
	$= (200 - t)^{-3}$	**A1**	oe, eg $\frac{1}{(200-t)^3}$.
	$x(200 - t)^{-3} = \int 12(200 - t)^{-3}\,\mathrm{d}t = \ldots$	**dM1**	Multiplies through by their integrating factor, LHS as $x\times$ their factor, and integrates the RHS with the power of $(200 - t)$ increased by 1.
	$x = 6(200 - t) + c(200 - t)^3$	**A1**	oe, including $x(200 - t)^{-3} = 6(200 - t)^{-2} + c$. Constant required.
	$t = 0,\ x = 0$ $\Rightarrow 0 = 1200 + 200^3 c \Rightarrow c = \ldots$	**M1**	Substitutes $t = 0$, $x = 0$ into their general solution containing a constant and solves for it.
	$x = 6(200 - t) - \frac{3}{20000}(200 - t)^3$	**A1**	oe.
	$\dfrac{\text{"}x\text{"}}{3(200 - t)} = 1.5$ $\Rightarrow (200 - t)^2 = 10000 \Rightarrow t = \ldots$	**M1**	Sets their x divided by their volume equal to 1.5, or their x equal to 1.5 times their volume, and solves for t.
	$t = 100$ (minutes)	**A1**	A0 if $t = 300$ is also given as an answer.
		[8]	

Question 3 continued

Question	Scheme	Marks	Notes
3(c)	Allow the concentration to vary within the tank (non-uniform mixing)	**E1**	States a refinement changing a modelling assumption, eg non-uniform mixing, time to dissolve, variable flow rates. Not for vague accuracy comments.
		[1]	
	Total for Question 3	**13**	

Question	Scheme	Marks	Notes
4(a)	(i) $V = 15$ litres	**B1**	Or 15000 ml.
	(ii) Rate in $= 0.2(4 + e^{-0.1t})$ $= 0.8 + 0.2e^{-0.1t}$	**B1**	oe, such as $\frac{200}{1000}(4 + e^{-0.1t})$.
	Rate out $= 0.3 \times \frac{m}{15}$	**M1**	Forms rate out as the outflow rate times m divided by their constant volume from (i), in consistent units.
	$\frac{\mathrm{d}m}{\mathrm{d}t} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50}$	**A1 AG**	Rate in minus rate out, both correct, leading without error to the printed equation.
		[4]	
4(b)	$e^{\int \frac{1}{50}\,\mathrm{d}t} = e^{t/50}$	**B1**	oe, such as $e^{0.02t}$.
	$me^{t/50} = \int \left(0.8e^{t/50} + 0.2e^{-0.08t}\right) \mathrm{d}t$ $= \ldots$	**M1**	Writes LHS as m times their integrating factor of the form e^{kt}, multiplies the RHS by it and integrates at least one term to the form pe^{qt}.
	$me^{t/50} = 40e^{t/50} - 2.5e^{-0.08t} + c$	**A1**	oe, including $m = 40 - 2.5e^{-0.1t} + ce^{-0.02t}$. Constant required.
	$t = 0,\ m = 6 \Rightarrow 6 = 40 - 2.5 + c \Rightarrow c = \ldots$	**M1**	Substitutes $t = 0$, $m = 6$ into their general solution containing a constant and solves for it.
	$m = 40 - 2.5e^{-0.1t} - 31.5e^{-0.02t}$	**A1**	oe, with m as the subject.
		[5]	
4(c)	$\frac{m(30)}{15} = \frac{40 - 2.5e^{-3} - 31.5e^{-0.6}}{15} = \ldots$	**M1**	Substitutes $t = 30$ into their m and divides by their volume from (a)(i).
	$= 1.51$, close to 1.52, so the model is good	**A1ft**	ft their m: their value (awrt 1.51 if correct) compared with 1.52, concluding the model is good if within 1.44 to 1.60, poor otherwise.
		[2]	
	Total for Question 4	**11**	

Question	Scheme	Marks	Notes
5(a)	$2\frac{\mathrm{d}v}{\mathrm{d}t} = -2t(v+3)$ $\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + tv = -3t$	**M1**	Uses $F = ma$ with mass 2 and acceleration $\frac{\mathrm{d}v}{\mathrm{d}t}$, and rearranges to the given form. Condone sign slips.
	$P(t) = t,\ Q(t) = -3t$	**A1**	Both correct.
		[2]	
5(b)	$e^{\int t\,\mathrm{d}t} = e^{t^2/2}$	**B1**	
	$ve^{t^2/2} = \int -3te^{t^2/2}\,\mathrm{d}t$ $= -3e^{t^2/2} + c$	**M1 A1**	**M1** Multiplies through by their integrating factor of the form $e^{f(t)}$ and integrates both sides, RHS to the form $ke^{t^2/2}$. Condone sign and coefficient slips. **A1** Correct, with constant, oe such as $v = -3 + ce^{-t^2/2}$.
	$5 = -3 + c \Rightarrow c = \ldots$	**M1**	Substitutes $t = 0$, $v = 5$ into their general solution containing a constant to find it.
	$v = -3 + 8e^{-t^2/2}$	**A1**	oe, with v the subject.
	$8e^{-t^2/2} = 3 \Rightarrow -\frac{t^2}{2} = \ln\frac{3}{8}$ $\Rightarrow t^2 = \ldots$	**M1**	Sets their v of the form $a + be^{-t^2/2}$ to zero and takes logarithms correctly to reach $t^2 = \ldots$ or $t = \ldots$.
	$t = \sqrt{2\ln\frac{8}{3}}$ $[\approx 1.40]$	**A1**	Exact form or awrt 1.40, positive value only.
		[7]	
	Total for Question 5	**9**	

Question	Scheme	Marks	Notes
6(a)	$e^{\int(1-\cot t)\,\mathrm{d}t} = e^{t-\ln(\sin t)}$	**M1**	Uses $e^{\int(1-\cot t)\,\mathrm{d}t}$ and integrates at least one term correctly.
	$= \frac{e^t}{\sin t}$	**A1**	oe, e.g. $e^t \operatorname{cosec} t$.
	$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{e^t}{\sin t}R\right) = \frac{1}{1+t^2}$ $\Rightarrow \frac{e^t}{\sin t}R = \int \frac{1}{1+t^2}\,\mathrm{d}t = \ldots$	**dM1**	Multiplies through by their integrating factor, takes LHS as their factor $\times R$, and integrates RHS to the form $k\arctan t$. Constant may be absent.
	$\frac{e^t}{\sin t}R = \arctan t + c$	**A1**	Must include the constant; R need not be the subject.
	$\pi e^{-1}\sin 1 = e^{-1}\sin 1\,(\arctan 1 + c)$ $\Rightarrow c = \ldots$	**M1**	Substitutes $t = 1$, $R = \pi e^{-1}\sin 1$ into their general solution containing a constant and solves for it.
	$\pi = \frac{\pi}{4} + c \Rightarrow c = \frac{3\pi}{4}$ $R = e^{-t}\sin t\left(\arctan t + \frac{3\pi}{4}\right)$	**A1 AG**	cso. Correct substituted equation with $\arctan 1 = \frac{\pi}{4}$, leading without error to the printed R.
		[6]	
6(b)	$R = 0$, $e^{-t} > 0$, $\arctan t + \frac{3\pi}{4} > 0$ $\Rightarrow \sin t = 0$	**M1**	Sets $R = 0$ and reduces to $\sin t = 0$. No justification that the other factors are nonzero is needed.
	$t = \pi$ seconds	**A1**	Or awrt 3.14. A0 for $t = n\pi$ without selecting π.
		[2]	
6(c)	$\frac{\mathrm{d}R}{\mathrm{d}t} = -e^{-t}\sin t\left(\arctan t + \frac{3\pi}{4}\right)$ $+ e^{-t}\cos t\left(\arctan t + \frac{3\pi}{4}\right)$ $+ \frac{e^{-t}\sin t}{1+t^2}$	**M1 A1**	**M1** Uses the product rule, giving a sum of terms each with one factor differentiated. Condone sign and coefficient slips. **A1** oe; need not be simplified.
	$t = 0 \Rightarrow \frac{\mathrm{d}R}{\mathrm{d}t} = \frac{3\pi}{4}$	**A1**	
		[3]	
	Total for Question 6	**11**	

Question	Scheme	Marks	Notes
7(a)	$\frac{\mathrm{d}V}{\mathrm{d}t} - \frac{3-2t}{3t-t^2}V = \frac{3t-t^2}{1+(t-1)^2}$ $e^{\int -\frac{3-2t}{3t-t^2}\,\mathrm{d}t} = \ldots$	**M1**	Divides by $3t-t^2$ and uses $e^{\int P\,\mathrm{d}t}$ with their P. Condone a sign slip in P.
	$e^{-\ln(3t-t^2)} = \frac{1}{3t-t^2}$	**A1**	
	$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{V}{3t-t^2}\right) = \frac{1}{1+(t-1)^2}$ $\Rightarrow \frac{V}{3t-t^2} = \ldots$	**dM1**	Multiplies through by their integrating factor, writes the LHS as the derivative of $V\times$ it, and integrates the RHS to a form $k\arctan(t-1)$.
	$\frac{V}{3t-t^2} = \arctan(t-1) + c$	**A1**	
	$\frac{4}{3-1} = \arctan 0 + c \Rightarrow c = \ldots$	**M1**	Substitutes $t=1$, $V=4$ into their general solution containing a constant and solves for the constant.
	$c = 2$ $\Rightarrow V = (3t-t^2)\left(\arctan(t-1)+2\right)$	**A1 AG**	Correct substituted condition giving $c=2$, leading without error to the printed result.
		[6]	
7(b)	$(3t-t^2)\left(\arctan(t-1)+2\right) = 0$ $\arctan(t-1)+2 > 2-\frac{\pi}{2} > 0$ $\Rightarrow 3t-t^2 = 0 \Rightarrow t = \ldots$	**M1**	Sets $V=0$ and solves $3t-t^2=0$. No justification that $\arctan(t-1)+2 \neq 0$ is needed.
	$t=3$: the pulse lasts 3 seconds	**A1**	3 alone selected as the length; A0 if $t=0$ is also given as an answer.
		[2]	
7(c)	$\frac{\mathrm{d}V}{\mathrm{d}t} = (3-2t)\left(\arctan(t-1)+2\right)$ $\quad + \frac{3t-t^2}{1+(t-1)^2}$	**M1 A1**	**M1** Uses the product rule: a sum of two products, each with one factor differentiated. Condone slips in the derivatives.
	$t=0$: $3\left(\arctan(-1)+2\right) = 6-\frac{3\pi}{4}$	**A1**	oe exact, such as $3\left(2-\frac{\pi}{4}\right)$.
		[3]	
Total for Question 7		**11**	

Question	Scheme	Marks	Notes
8(a)(i)	$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC$	**B1**	oe, with the sign showing a decrease; any letter for the constant.
		[1]	
8(a)(ii)	$\frac{\mathrm{d}C}{\mathrm{d}t} = -kAe^{-kt}$	**M1**	Differentiates to the form pAe^{-kt} and substitutes into, or compares with, their equation from (a)(i).
	$= -kC$	**A1**	Correct derivative shown equal to $-kC$, with a statement that the equation is satisfied.
	$C(0) = Ae^0 = A$ $k > 0 \Rightarrow e^{-kt} \to 0 \Rightarrow C \to 0$	**B1**	Both conditions shown.
		[3]	
8(b)	$e^{\int \frac{2t}{1+t^2}\,\mathrm{d}t} = e^{\ln(1+t^2)}$	**M1**	Uses $e^{\int P\,\mathrm{d}t}$ and integrates to the form $\lambda\ln(1+t^2)$.
	$= 1 + t^2$	**A1 AG**	From $e^{\ln(1+t^2)}$ without error.
		[2]	
8(c)(i)	$\frac{\mathrm{d}}{\mathrm{d}t}\left((1+t^2)C\right) = 0$ $\Rightarrow (1+t^2)C = K$	**M1 A1**	**M1** Multiplies by $1+t^2$ and integrates, LHS to $(1+t^2)C$; or separates variables and integrates both sides. **A1** oe, with a constant.
	$t = 0,\ C = A \Rightarrow K = \ldots\ [= A]$	**M1**	Substitutes $t = 0$, $C = A$ into their solution with a constant to find it.
	$C = \frac{A}{1+t^2}$	**A1 AG**	From a correct general solution and $K = A$, without error.
		[4]	
8(c)(ii)	$\frac{A}{1+t^2} = \frac{A}{2} \Rightarrow t = \ldots$	**M1**	Sets the given expression equal to $\frac{A}{2}$ and solves for t.
	$t = 1$ hour $= 60$ minutes	**A1**	60 minutes or 1 hour, with no negative value given as the answer.
		[2]	
8(d)	$(1+t^2)\frac{\mathrm{d}C}{\mathrm{d}t} + 2tC = te^{-t}$ $\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left((1+t^2)C\right) = te^{-t}$	**M1 A1**	**M1** Multiplies through by $1+t^2$; RHS need not be simplified. **A1** LHS as the derivative of $(1+t^2)C$, or implied by integrating to $(1+t^2)C$.

Question 8 continued

Question	Scheme	Marks	Notes
	$\int te^{-t}\,\mathrm{d}t = -te^{-t} + \int e^{-t}\,\mathrm{d}t$	**M1**	Integrates by parts with $u = t$, reaching the form $\pm te^{-t} \pm \int e^{-t}\,\mathrm{d}t$. Condone sign slips.
	$(1+t^2)C = -(1+t)e^{-t} + K$	**A1**	oe, with a constant.
	$t = 0,\ C = A \Rightarrow A = -1 + K$ $\Rightarrow K = \ldots$	**M1**	Substitutes $t = 0$, $C = A$ into their solution with a constant to find it.
	$C = \frac{A+1-(1+t)e^{-t}}{1+t^2}$	**A1 AG**	From a correct general solution and $K = A + 1$, without error.
		[6]	
8(e)	$\frac{6-(1+t)e^{-t}}{1+t^2} = 2.5 \Rightarrow t = \ldots$	**M1**	Sets model (d) with $A = 5$ equal to 2.5 and solves for t; calculator or numerical method accepted.
	$t = 1.055\ldots$ hours ≈ 63 minutes Closer to 63 than 60, so model (d)	**A1ft**	Awrt 63 minutes or 1.06 hours, compared with 63 and their (c)(ii) time, choosing the closer model. ft their (c)(ii).
		[2]	
	Total for Question 8	**20**	

Question	Scheme	Marks	Notes
9(a)	$300 - 200 = 100$ ml $\text{min}^{-1} = 0.1$ L min^{-1} $\Rightarrow V = 4 + 0.1t$	**B1**	Net rate of 100 ml or 0.1 litres per minute seen; not for the printed volume alone.
	Rate in $= 0.3 \times 2e^{-0.1t} = 0.6e^{-0.1t}$	**B1**	
	Rate out $= 0.2 \times \dfrac{m}{4 + 0.1t}$	**M1**	Multiplies the outflow rate by $\frac{m}{V}$, with their volume. Condone inconsistent units (200 for 0.2).
	$\dfrac{\mathrm{d}m}{\mathrm{d}t} = 0.6e^{-0.1t} - \dfrac{0.2m}{4 + 0.1t}$ $= 0.6e^{-0.1t} - \dfrac{2m}{40 + t}$	**A1 AG**	Rate in minus rate out in consistent units, simplified without error to the printed equation.
		[4]	
9(b)	$\mu = e^{\int \frac{2}{40+t}\,\mathrm{d}t}$	**M1**	Forms $e^{\int P\,\mathrm{d}t}$ with $P = \frac{2}{40+t}$ from the standard form. Condone a sign slip in P.
	$= (40 + t)^2$	**A1**	
	$(40 + t)^2 m = \int 0.6(40 + t)^2 e^{-0.1t}\,\mathrm{d}t$ $= \ldots$	**dM1**	Multiplies through by their integrating factor μ of the form $(40 + t)^n$, $n \geq 1$, integrates LHS to μm, and applies parts to RHS, differentiating the polynomial factor. Condone coefficient slips.
	$= -6(t^2 + 80t + 1600)e^{-0.1t}$ $+ 12\int (t + 40)e^{-0.1t}\,\mathrm{d}t$	**A1**	oe, $(40 + t)^2$ expanded or not.
	$\int (t + 40)e^{-0.1t}\,\mathrm{d}t$ $= -10(t + 40)e^{-0.1t} + \int 10e^{-0.1t}\,\mathrm{d}t$	**dM1**	Applies parts to the remaining integral, differentiating the linear factor. Condone coefficient slips.
	$\int 0.6(40 + t)^2 e^{-0.1t}\,\mathrm{d}t$ $= -6e^{-0.1t}(t^2 + 100t + 2600)$	**A1**	oe.
	$(40 + t)^2 m = -6e^{-0.1t}(t^2 + 100t + 2600) + c$	**A1**	oe; constant of integration required.
	$40^2 \times 4 = -6(2600) + c \Rightarrow c = \ldots$ $[= 22000]$	**M1**	Substitutes $t = 0$, $m = 4$ into their solution containing a constant of integration and solves for it.
	$m = \dfrac{22000 - 6e^{-0.1t}(t^2 + 100t + 2600)}{(40 + t)^2}$	**A1**	oe, with m as subject.
		[9]	

Question 9 continued

Question	Scheme	Marks	Notes
9(c)	$m(20) = \dfrac{22000 - 30000e^{-2}}{3600}$, $V = 6$ $\dfrac{m(20)}{6} = \ldots$	**M1**	Substitutes $t = 20$ into their m and divides by their volume at $t = 20$.
	≈ 0.831, close to 0.82, so the model is good	**A1ft**	awrt 0.831, ft their m. Concludes model good if within 5% of 0.82, otherwise not, with comparison stated.
		[2]	
	Total for Question 9	**15**	

Question	Scheme	Marks	Notes
10(a)	Resistance increases with speed, and is zero when the boat is at rest	**E1**	States that resistance increases with, or depends on, speed, or is zero at rest. Not for "more realistic" alone.
		[1]	
10(b)	$m\frac{\mathrm{d}v}{\mathrm{d}t} = -0.2mv - m$ $\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = -1$	**B1**	Both resistances opposing the motion in $F = ma$, then division by m to the printed result.
		[1]	
10(c)(i)	$e^{\int 0.2\,\mathrm{d}t} = e^{t/5}$	**B1**	
	$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/5}\right) = -e^{t/5}$ $\Rightarrow ve^{t/5} = \int -e^{t/5}\,\mathrm{d}t$	**M1**	Multiplies through by their integrating factor e^{kt} and integrates both sides, the RHS to the form pe^{kt}. Separating variables accepted.
	$ve^{t/5} = -5e^{t/5} + c$ $\Rightarrow v = -5 + ce^{-t/5}$	**A1**	oe, in either form, with the constant.
	$0 = -5 + ce^{-2} \Rightarrow c = \ldots$	**M1**	Substitutes $t = 10$, $v = 0$ into their general solution containing a constant and solves for it.
	$v = 5\left(e^{2-t/5} - 1\right)$	**A1**	oe, eg $5e^2e^{-t/5} - 5$, or with c awrt 36.9.
		[5]	
10(c)(ii)	$v = 5(e-1)$ $[\approx 8.59]$	**B1ft**	ft their (c)(i) solution at $t = 5$. Accept $5(e-1)$ or awrt 8.59.
		[1]	
10(d)	$m\frac{\mathrm{d}v}{\mathrm{d}t} = kt^2 - 0.2mv$ $\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = \lambda t^2,\ \lambda = \frac{k}{m} > 0$	**B1**	Driving force kt^2 and opposing resistance $0.2mv$ in $F = ma$, division by m, and λ identified as a positive constant.
		[1]	
10(e)(i)	$e^{\int 0.2\,\mathrm{d}t} = e^{t/5}$	**B1**	
	$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/5}\right) = \lambda t^2 e^{t/5}$ $\int t^2e^{t/5}\,\mathrm{d}t = 5t^2e^{t/5} - \int 10te^{t/5}\,\mathrm{d}t$	**M1 A1**	**M1** Multiplies through by their e^{kt} and applies parts to $\int t^2e^{kt}\,\mathrm{d}t$, reaching the form $pt^2e^{kt} - \int qte^{kt}\,\mathrm{d}t$. Condone coefficient slips. **A1** Correct, with or without λ.

Question 10 continued

Question	Scheme	Marks	Notes
	$\int te^{t/5}\,\mathrm{d}t = 5te^{t/5} - \int 5e^{t/5}\,\mathrm{d}t$	**dM1**	Applies parts again to their $\int te^{kt}\,\mathrm{d}t$, reaching the form $pte^{kt} \pm qe^{kt}$; a tabular method accepted. Condone coefficient slips.
	$\int t^2e^{t/5}\,\mathrm{d}t = 5e^{t/5}\left(t^2 - 10t + 50\right)$	**A1**	oe, eg expanded.
	$v = 5\lambda\left(t^2 - 10t + 50\right) + ce^{-t/5}$	**A1**	oe, eg as $ve^{t/5} = \ldots$, with the constant.
	$0 = 250\lambda + c \Rightarrow c = \ldots$	**M1**	Substitutes $t = 0$, $v = 0$ into their general solution containing a constant and solves for it.
	$c = -250\lambda$ $\Rightarrow v = 5\lambda\left(t^2 - 10t + 50 - 50e^{-t/5}\right)$	**A1 AG**	$c = -250\lambda$ from a correct substituted equation, leading without error to the printed result. A0 if λ is dropped and reinstated.
		[8]	
10(e)(ii)	$5\lambda\left(25 - 50e^{-1}\right) = "5(e-1)"$ $\Rightarrow \lambda = \ldots$	**M1**	Substitutes $t = 5$ in the printed expression and equates to their (c)(ii) velocity.
	$\lambda = \frac{e(e-1)}{25(e-2)}$ $[\approx 0.260]$	**A1**	oe, or awrt 0.260.
		[2]	
10(f)	$\int_0^5 5\lambda\left(t^2 - 10t + 50 - 50e^{-t/5}\right)\mathrm{d}t$ $= 5\lambda\left[\frac{t^3}{3} - 5t^2 + 50t + 250e^{-t/5}\right]_0^5$	**M1 A1**	**M1** Integrates the (e)(i) expression: polynomial powers increased by 1 and $e^{-t/5} \to ke^{-t/5}$. Condone coefficient slips. **A1** Correct, in λ or its value.
	$\int_5^{10} 5\left(e^{2-t/5} - 1\right)\mathrm{d}t$ $= 5\left[-5e^{2-t/5} - t\right]_5^{10}$	**M1 A1**	**M1** Integrates their (c)(i) expression: $e^{2-t/5} \to ke^{2-t/5}$ and the constant to a multiple of t.
	$s = 5\lambda\left(\frac{250}{e} - \frac{250}{3}\right) + 25(e-2)$ $= \ldots$ $[\approx 11.2 + 18.0]$	**dM1**	Applies limits 0 to 5 and 5 to 10 to their integrals, with their λ, and adds. Needs both previous M1 marks.
	29 m	**A1**	
		[6]	
	Total for Question 10	**25**	

Question	Scheme	Marks	Notes
11(a)	Resistance increases as the speed of the cage increases	**E1**	States that resistance increases with, or depends on, speed. Not for "more realistic" alone.
		[1]	
11(b)	$m\frac{\mathrm{d}v}{\mathrm{d}t} = 0.75mg - mg - 0.25mv$ $\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = -\frac{1}{4}g$	**B1**	Newton's second law with tension, weight and resistance, correct signs, leading without error to the printed equation.
		[1]	
11(c)(i)	$e^{\int 0.25\,\mathrm{d}t} = e^{t/4}$	**B1**	
	$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/4}\right) = -\frac{1}{4}ge^{t/4}$ $\Rightarrow ve^{t/4} = \ldots$	**M1**	Multiplies through by their integrating factor and integrates both sides, LHS to v times their integrating factor.
	$ve^{t/4} = -ge^{t/4} + c$	**A1**	oe; needs the constant.
	$0 = ce^{-2} - g \Rightarrow c = \ldots$	**M1**	Substitutes $t = 8$, $v = 0$ into their solution containing a constant and solves for it.
	$v = g\left(e^{2-t/4} - 1\right)$	**A1**	oe, with g or 9.8.
		[5]	
11(c)(ii)	$v = g(e-1)$ $[\approx 16.8]$	**B1ft**	ft their (c)(i) solution at $t = 4$, exact or to 3 s.f.
		[1]	
11(d)	$m\frac{\mathrm{d}v}{\mathrm{d}t} = m(g + \lambda t) - mg - 0.25mv$ $\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = \lambda t$	**B1**	Newton's second law with tension, weight and resistance, correct signs, leading without error to the printed equation.
		[1]	
11(e)(i)	$e^{\int 0.25\,\mathrm{d}t} = e^{t/4}$	**B1**	
	$ve^{t/4} = \lambda \int te^{t/4}\,\mathrm{d}t$ $\quad = \lambda\left(4te^{t/4} - \int 4e^{t/4}\,\mathrm{d}t\right)$	**M1**	Multiplies through by their integrating factor and integrates $te^{t/4}$ by parts to the form $pte^{t/4} - \int qe^{t/4}\,\mathrm{d}t$. Condone sign and coefficient slips.
	$\int te^{t/4}\,\mathrm{d}t = 4te^{t/4} - 16e^{t/4}$	**A1**	
	$v = 4\lambda(t-4) + ce^{-t/4}$	**A1**	oe, such as $ve^{t/4} = \lambda\left(4te^{t/4} - 16e^{t/4}\right) + c$; needs the constant.

Question 11 continued

Question	Scheme	Marks	Notes
	$0 = -16\lambda + c \Rightarrow c = \ldots$	**M1**	Substitutes $t = 0$, $v = 0$ into their solution containing a constant and solves for it.
	$c = 16\lambda$ $\Rightarrow v = 4\lambda\left(t - 4 + 4e^{-t/4}\right)$	**A1 AG**	Correct c substituted into a correct general solution, leading without error to the printed expression.
		[6]	
11(e)(ii)	$4\lambda\left(4 - 4 + 4e^{-1}\right) = "g(e-1)"$ $\Rightarrow \lambda = \ldots$	**M1**	Equates $v(4)$ from the printed expression to their (c)(ii) velocity and solves for λ.
	$\lambda = \frac{ge(e-1)}{16}$ $[\approx 2.86]$	**A1**	oe, or awrt 2.86.
		[2]	
11(f)	$\int 4\lambda\left(t - 4 + 4e^{-t/4}\right)\mathrm{d}t$ $= 4\lambda\left[\frac{t^2}{2} - 4t - 16e^{-t/4}\right]$	**M1 A1**	**M1** Integrates the first-stage v: power of t increased in at least one term, or $e^{-t/4} \to ke^{-t/4}$. **A1** oe; a correct definite value $\left(32 - \frac{64}{e}\right)\lambda$ implies it.
	$\int g\left(e^{2-t/4} - 1\right)\mathrm{d}t$ $= g\left[-4e^{2-t/4} - t\right]$	**M1 A1**	**M1** Integrates their second-stage v, of the form $Ae^{-t/4} + B$, with $e^{-t/4} \to ke^{-t/4}$ or the constant to a multiple of t. **A1** oe; a correct definite value $4g(e-2)$ implies it.
	$\left(32 - \frac{64}{e}\right)"\lambda" + 4g(e-2) = \ldots$	**dM1**	Applies limits 0, 4 and 4, 8 to their integrals and adds, with their λ.
	52 (m)	**A1**	
		[6]	
	Total for Question 11	**23**	

Question	Scheme	Marks	Notes
12(a)(i)	Volume $= 2 + rt - st$	**M1**	Forms the volume as initial volume plus inflow minus outflow in time t.
	$= 2 + (r-s)t$ $\Rightarrow$ concentration $= \frac{m}{2+(r-s)t}$	**A1 AG**	Mass divided by the correct volume, reaching the printed result.
		[2]	
12(a)(ii)	$\frac{\mathrm{d}m}{\mathrm{d}t} = 10r - s \times \frac{m}{2+(r-s)t}$	**M1**	Forms $\frac{\mathrm{d}m}{\mathrm{d}t}$ as rate in $10r$ minus rate out s times the concentration from (i). Condone a sign slip.
	$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{sm}{2+(r-s)t} = 10r$	**A1 AG**	Correct equation rearranged to the printed result without error.
		[2]	
12(b)(i)	$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{r}{2}m = 10r$ $\mu = e^{rt/2}$	**B1**	
	$\frac{\mathrm{d}}{\mathrm{d}t}\left(me^{rt/2}\right) = 10re^{rt/2}$ $\Rightarrow me^{rt/2} = \ldots$	**M1**	Multiplies through by their integrating factor of the form e^{kt} and integrates the RHS to the form Ae^{kt}.
	$me^{rt/2} = 20e^{rt/2} + c$	**A1**	oe, with the constant.
	$4 = 20 + c \Rightarrow c = \ldots$	**M1**	Substitutes $t = 0$, $m = 4$ into their general solution containing a constant and finds it.
	$m = 20 - 16e^{-rt/2}$	**A1**	oe, m in terms of r and t.
		[5]	
12(b)(ii)	$\frac{\text{"}20-16e^{-r}\text{"}}{2} = 8 \Rightarrow e^{-r} = \ldots$ $\Rightarrow r = \ldots$	**M1**	Sets their $\frac{m}{2}$ equal to 8 at $t = 2$, with m of the form $a + be^{-kt}$, and solves using logarithms. Not for setting $m = 8$.
	$r = \ln 4$	**A1**	oe, or awrt 1.39.
		[2]	
12(c)(i)	Volume $= 2 - rt = 0$ when $t = \frac{2}{r}$: tank empty	**E1**	Uses the volume $2 - rt$ to conclude the tank is empty at $t = \frac{2}{r}$. A zero denominator alone, without reference to volume, is not enough.
		[1]	

Question 12 continued

Question	Scheme	Marks	Notes
12(c)(ii)	$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{2r}{2-rt}m = 10r$ $\mu = e^{\int \frac{2r}{2-rt}\,\mathrm{d}t}$	**M1**	Substitutes $s = 2r$ and writes the integrating factor as $e^{\int P\,\mathrm{d}t}$ with their P of the form $\frac{k}{2-rt}$.
	$= e^{-2\ln(2-rt)} = (2-rt)^{-2}$	**A1**	Simplified from the exponential; accept $\frac{1}{(2-rt)^2}$.
	$\frac{\mathrm{d}}{\mathrm{d}t}\left(m(2-rt)^{-2}\right) = 10r(2-rt)^{-2}$ $\Rightarrow m(2-rt)^{-2} = \ldots$	**dM1**	Multiplies through by their integrating factor of the form $(2-rt)^n$ and integrates the RHS to the form $A(2-rt)^{n+1}$.
	$m(2-rt)^{-2} = \frac{10}{2-rt} + c$	**A1**	oe, with the constant.
	$4 = 20 + 4c \Rightarrow c = \ldots$	**M1**	Substitutes $t = 0$, $m = 4$ into their general solution containing a constant and finds it.
	$m = 10(2-rt) - 4(2-rt)^2$ $[= 4 + 6rt - 4r^2t^2]$	**A1**	oe.
	$\frac{\mathrm{d}m}{\mathrm{d}t} = 6r - 8r^2t = 0 \Rightarrow t = \ldots$	**M1**	Differentiates their m, at least one power of t reduced by 1, sets it to zero and solves for t; completing the square equally valid.
	$t = \frac{3}{4r}$	**A1**	No justification of the maximum or of $t < \frac{2}{r}$ needed.
	$m = 4 + 6r\left(\frac{3}{4r}\right) - 4r^2\left(\frac{3}{4r}\right)^2$	**dM1**	Substitutes their t into their particular solution.
	$m_{\max} = \frac{25}{4}$ (g)	**A1**	Accept 6.25.
		[10]	
	Total for Question 12	**22**	

Question	Scheme	Marks	Notes
13(a)(i)	Volume $= 4 + pt - qt$	**M1**	Forms the volume at time t as initial volume plus inflow minus outflow.
	Proportion $= \dfrac{x}{4 + (p-q)t}$	**A1 AG**	Divides x by the correct volume and reaches the printed form without error.
		[2]	
13(a)(ii)	$\frac{\mathrm{d}x}{\mathrm{d}t} = 0.6p - q \times \frac{x}{4+(p-q)t}$	**M1**	Forms $\frac{\mathrm{d}x}{\mathrm{d}t}$ as rate in minus rate out, with rate out as q times the proportion from (a)(i). Condone sign slips.
	$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{qx}{4+(p-q)t} = \frac{3p}{5}$	**A1 AG**	Correct rate equation, with rate in $0.6p$ or $\frac{3p}{5}$, rearranged to the printed result without error.
		[2]	
13(b)(i)	$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{p}{4}x = \frac{3p}{5}$ $\mu = e^{pt/4}$	**B1**	
	$xe^{pt/4} = \int \frac{3p}{5} e^{pt/4}\,\mathrm{d}t$	**M1**	Multiplies through by their integrating factor e^{kt} and integrates both sides, RHS to the form Ae^{kt}; or separates the variables and integrates.
	$xe^{pt/4} = \frac{12}{5}e^{pt/4} + c$	**A1**	Correct, including the constant; any equivalent form.
	$t = 0,\ x = 0 \Rightarrow c = \ldots\ [= -\frac{12}{5}]$	**M1**	Substitutes $t = 0$, $x = 0$ into their general solution containing a constant to find it.
	$x = \frac{12}{5}\left(1 - e^{-pt/4}\right)$	**A1**	oe, with x the subject.
		[5]	
13(b)(ii)	$0.3 \times 4 =$ "$\frac{12}{5}\left(1 - e^{-p}\right)$" $\Rightarrow e^{-p} = \ldots \Rightarrow p = \ldots$	**M1**	Equates 0.3×4 to their x, of the form $A(1 - e^{-kpt})$, at $t = 4$ and uses logarithms to solve for p.
	$p = \ln 2$	**A1**	Or awrt 0.693.
		[2]	
13(c)(i)	Volume $= 4 - pt = 0$ at $t = \frac{4}{p}$: tank empty	**E1**	Shows the volume $4 - pt$ is zero at $t = \frac{4}{p}$ and concludes the tank is empty, or that the denominator is zero or negative for $t \geq \frac{4}{p}$.
		[1]	

Question 13 continued

Question	Scheme	Marks	Notes
13(c)(ii)	$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2px}{4-pt} = \frac{3p}{5}$ $\mu = e^{\int \frac{2p}{4-pt}\,\mathrm{d}t}$	**M1**	Uses $q = 2p$ and writes the integrating factor as e to the integral of their coefficient of x, integrated to a logarithm.
	$= e^{-2\ln(4-pt)} = (4-pt)^{-2}$	**A1**	oe such as $\frac{1}{(4-pt)^2}$.
	$x(4-pt)^{-2} = \int \frac{3p}{5}(4-pt)^{-2}\,\mathrm{d}t$	**dM1**	Multiplies through by their integrating factor and integrates both sides, RHS to the form $k(4-pt)^{-1}$.
	$x(4-pt)^{-2} = \frac{3}{5(4-pt)} + c$	**A1**	Correct, including the constant; any equivalent form.
	$t = 0,\ x = 0 \Rightarrow c = \ldots\ [= -\frac{3}{20}]$	**M1**	Substitutes $t = 0$, $x = 0$ into their general solution containing a constant to find it.
	$x = \frac{3}{5}(4-pt) - \frac{3}{20}(4-pt)^2$ $[= \frac{3pt}{5} - \frac{3p^2t^2}{20}]$	**A1**	oe, with x the subject; need not be expanded.
	$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{3p}{5} - \frac{3p^2t}{10} = 0$ $\Rightarrow t = \ldots$	**M1**	Differentiates their quadratic x, power of at least one term reduced by 1, and sets it to zero; or completes the square.
	$t = \frac{2}{p}$	**A1**	
	$x = \frac{3p}{5} \times \frac{2}{p} - \frac{3p^2}{20} \times \frac{4}{p^2}$	**dM1**	Substitutes their t into their particular solution.
	Maximum $= \frac{3}{5}$ litres	**A1**	oe such as 0.6. No justification that it is a maximum is needed.
		[10]	
Total for Question 13		**22**	

Question	Scheme	Marks	Notes
14(a)(i)	$\frac{\mathrm{d}C}{\mathrm{d}t} = k(M - C)$	**B1**	Any letter for the constant. A proportionality statement without an equation scores B0.
		[1]	
14(a)(ii)	$\frac{\mathrm{d}C}{\mathrm{d}t} = Mke^{-kt}$ $k(M - C) = k\left(M - M(1 - e^{-kt})\right)$	**M1**	Differentiates C to the form αe^{-kt} and substitutes C into $k(M - C)$, or into their equation from (a)(i).
	$= kMe^{-kt} = \frac{\mathrm{d}C}{\mathrm{d}t}$	**A1**	cso. Both sides shown equal, with a conclusion.
	$C(0) = M(1 - 1) = 0$ $e^{-kt} \to 0 \Rightarrow C \to M$ as $t \to \infty$	**B1**	Both conditions, with $e^{-kt} \to 0$ stated or implied.
		[3]	
14(b)	$-\frac{t+2}{t(1+t+t^2)} = \frac{A}{t} + \frac{Bt+D}{1+t+t^2}$	**M1**	Uses partial fractions of the form $\frac{A}{t} + \frac{Bt+D}{1+t+t^2}$; the fraction may be taken with either sign.
	$-(t + 2) = A(1 + t + t^2) + t(Bt + D)$ $\Rightarrow A = \ldots,\ B = \ldots,\ D = \ldots$	**dM1**	Multiplies through and compares coefficients or substitutes values to find the constants. Condone slips.
	$-\frac{2}{t} + \frac{2t+1}{1+t+t^2}$	**A1**	oe, including the positive fraction with all signs reversed.
	$\int \ldots \mathrm{d}t = -2\ln t + \ln(1 + t + t^2)$ $\Rightarrow e^{-2\ln t + \ln(1+t+t^2)}$	**M1**	Integrates their partial fractions to $\alpha \ln t + \beta \ln(1 + t + t^2)$ and forms $e^{\int P\,\mathrm{d}t}$. Condone a sign slip in P.
	$= \frac{1+t+t^2}{t^2}$	**A1 AG**	Correct integral and log laws leading without error to the printed form; combined steps accepted.
		[5]	
14(c)(i)	$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1+t+t^2}{t^2}C\right) = 0$ $\Rightarrow \frac{1+t+t^2}{t^2}C = K$	**M1 A1**	**M1** Uses the integrating factor from (b), writing the LHS as $\frac{\mathrm{d}}{\mathrm{d}t}(\mu C)$, or separates variables, and integrates. **A1** Constant of integration required; oe such as $\ln C$ expression.
	$C = \frac{Kt^2}{1+t+t^2} \to K$ as $t \to \infty$ $\Rightarrow K = M$	**M1**	Rearranges for C and lets $t \to \infty$ to equate their constant to M.

Question 14 continued

Question	Scheme	Marks	Notes
	$C = \frac{Mt^2}{1+t+t^2}$	**A1 AG**	Correct limit with $K = M$ stated, leading without error to the printed result.
		[4]	
14(c)(ii)	$\frac{Mt^2}{1+t+t^2} = \frac{M}{2} \Rightarrow t^2 - t - 1 = 0$ $\Rightarrow t = \ldots$ $[= 1.618\ldots]$	**M1**	Sets $C = \frac{M}{2}$, forms a three-term quadratic and solves it by any means.
	97 minutes	**A1**	cao, oe such as 1 hour 37 minutes. Negative root rejected.
		[2]	
14(d)	$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1+t+t^2}{t^2}C\right) = \frac{1+t+t^2}{t^2} \times \frac{3t^2e^{-t}}{1+t+t^2}$ $= 3e^{-t}$	**M1 A1**	**M1** Multiplies through by the integrating factor, writing the LHS as $\frac{\mathrm{d}}{\mathrm{d}t}(\mu C)$ and the RHS as $\mu S(t)$.
	$\frac{1+t+t^2}{t^2}C = -3e^{-t} + K$ $\Rightarrow C = \frac{Kt^2 - 3t^2e^{-t}}{1+t+t^2}$	**dM1**	Integrates their RHS to the form $\alpha e^{-t} + K$ and divides by the integrating factor.
	$t^2e^{-t} \to 0 \Rightarrow C \to K$ as $t \to \infty$ $\Rightarrow K = M$	**M1**	Lets $t \to \infty$, with the exponential term tending to 0, to equate their constant to M.
	$C = \frac{Mt^2 - 3t^2e^{-t}}{1+t+t^2}$	**A1 AG**	Correct integration and limits, $t^2e^{-t} \to 0$ stated, leading without error to the printed result.
		[5]	
14(e)	$C(1.75) = \frac{10(1.75)^2 - 3(1.75)^2e^{-1.75}}{1+1.75+1.75^2}$ $= \ldots$ $[\approx 4.99]$	**M1**	Substitutes $M = 10$ and $t = 1.75$ hours into the model (d) expression, or solves $C = 5$ for t numerically.
	$4.99 \approx 5$; model (c) gives "97" min, not 105 $\Rightarrow$ model (d) more consistent	**A1ft**	ft their (c)(ii) time. Needs awrt 4.99 (or awrt 105 minutes) and a comparison showing model (d) closer to the data.
		[2]	
	Total for Question 14	**22**	